Exam Date & Time: 02-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

VII SEMESTER B.TECH END SEMESTER MAKEUP EXAMINATION, DEC 2024 - JAN 2025

Medical Imaging [BME 4058]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

Instructions to Candidates:
[i] Missing data may be suitably assumed
[ii] No formula book will be provided

1A)	Explain how the Photoelectric effect and Compton scattering impact the clarity of nuclear medicine images. What measures are taken to reduce their negative effects on image quality?	(3)
1B)	Analyze the process of applying the bilateral filter in preprocessing CT images. In what scenarios would this filter be preferable over traditional smoothing filters, especially in medical imaging?	(3)
1C)	Differentiate between the spatial and frequency domains in the context of CT filtering during the Filtered Back Projection (FBP) process. Why is filtering performed in the frequency domain before backprojection?	(4)
2A)	Compare the performance of SIRT and ART in terms of image quality and computational efficiency. Under what circumstances would SIRT be the preferred choice over ART for a large dataset?	(5)
2B)	Analyze how a gradient coil along the z-axis enables slice selection. What effect does changing the gradient field strength have on slice thickness?	(3)
2C)	Examine the role of shimming in MRI and how active versus passive shimming contributes to field homogeneity. Why is field homogeneity critical for image quality?	(2)
3A)	Analyze how adjustments to the RF pulse bandwidth and gradient field strength impact slice thickness. In which scenarios might a radiologist opt for a thinner slice versus a thicker slice?	(3)
3B)	Evaluate the consequences of increasing the frequency encoding gradient strength on the spatial resolution and potential artifacts in MRI images. Why might high gradient	(3)

strength lead to specific artifacts in certain body areas?

3C)	Analyze how increasing the number of phase encoding steps enhances image resolution along the y-axis but also increases scan duration. In what clinical imaging situations would the trade-off of increased scan time be justified?	(4)
4A)	Interpret how Fourier transformation enables spatial encoding in MRI. Why is Fourier transformation crucial in converting frequency and phase information to spatial locations?	(3)
4B)	Analyze the impact of each step in the MRI image reconstruction process, including slice selection, Fourier transformation, and windowing. How does each step contribute to creating an accurate and diagnostically useful image?	(4)
4C)	Interpret the data contained in the peripheral regions of a k-space image, where signals have lower amplitudes and higher frequencies. What type of spatial information is stored here, and how does it contribute to the MRI image's spatial resolution?	(3)
5A)	Analyze the relationship between ultrasound frequency, tissue attenuation, and penetration depth, and explain why higher frequencies provide better resolution.	(3)
5B)	Differentiate between electromagnetic waves and ultrasound waves.	(3)
5C)	Formulate an algorithm that details the sequence of steps involved in ultrasound image reconstruction starting from the acquisition of raw echo signals to the generation of a clinically interpretable image.	(4)

-----End-----